

Online Resource 1: Supporting Proofs, Methods, and Reproduction

for “Innovation-Safe Compression of Financial
Strategy Libraries: Semantics, Complexity, and
Algorithms”

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Abstract

This resource supplies the supporting probability and preservation arguments, covering and complexity proofs, algorithm guarantees, and deletion counterexample for the accompanying article. It also gives complete benchmark outcome accounting, financial and synthetic methods, origin-level financial results, verification scope, and instructions for reproducing the paper and its evidence. Public data are distributed separately in machine-readable form; licensed CRSP/WRDS observations require independent access.

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Evidence boundary: Human proof, Lean kernel verification, exact finite computation, solver evidence, synthetic evidence, and retrospective financial evidence are identified separately.

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S1 Exact probability normalization and projection identities

For a finite carrier A , every probability law in the model is an exact rational table

$$\Delta_{\mathbb{Q}}(A) := \left\{ p : A \rightarrow \mathbb{Q} : p(a) \geq 0 \text{ for every } a, \sum_{a \in A} p(a) = 1 \right\}.$$

Beliefs, belief-kernel rows, and candidate-generation rows use this convention. At fixed project q , belief b , and closure C , verification assigns

$$\Gamma(q, b, C)(s) = G(q, b, C)(s) \nu(q, b, C, s), \quad s \in S, \quad (\text{S1.1})$$

$$\Gamma(q, b, C)(\text{none}) = G(q, b, C)(\text{none}) + \sum_{s \in S} G(q, b, C)(s) (1 - \nu(q, b, C, s)). \quad (\text{S1.2})$$

Every term is nonnegative. Summing (S1.1)–(S1.2) cancels the verification split and recovers the unit mass of G , so Γ is normalized.

Conditional on initial belief b , productive state $K = (F, C)$, and project q of duration d_q , let $\Lambda_q(\mathbf{b}, o \mid b, K)$ be the exact joint law of the belief path $\mathbf{b} = (b_0, \dots, b_{d_q})$ and admitted outcome o . Its specified marginals are

$$\sum_o \Lambda_q(\mathbf{b}, o \mid b, K) = \mathbf{1}\{b_0 = b\} \prod_{t=0}^{d_q-1} P_B(b_t, b_{t+1}), \quad (\text{S1.3})$$

$$\sum_{\mathbf{b}} \Lambda_q(\mathbf{b}, o \mid b, K) = \Gamma(q, b, C)(o). \quad (\text{S1.4})$$

These equations do not impose independence between the belief path and the admitted outcome.

For a raw source library L , deterministic update gives

$$\mathcal{Q}_q^{\text{raw}}(b', L' \mid b, L) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ L \oplus o = L'}} \Lambda_q(\mathbf{b}, o \mid b, K_L).$$

For realizable $K = (F, C)$, the two compressed pushforwards are

$$\overline{T}_q(K' \mid b, K) := \sum_{o: \text{addK}(K, o) = K'} \Gamma(q, b, C)(o), \quad (\text{S1.5})$$

$$\overline{\mathcal{Q}}_q(b', K' \mid b, K) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ \text{addK}(K, o) = K'}} \Lambda_q(\mathbf{b}, o \mid b, K). \quad (\text{S1.6})$$

Finite deterministic pushforward preserves nonnegativity and total mass. The local update identity $K_{L \oplus o} = \text{addK}(K_L, o)$ also makes every compressed successor realizable. These identities supply the normalization clause of the raw-to-compressed projection theorem; they do not supply the factorization, positive-duration induction, or contraction arguments by themselves.

S1.1 Complete projection proof

The article assumes a finite hidden state, finite filtered belief carrier, finite strategy catalog, finite project catalog, positive integer project durations, exact rational transition and completion laws, and a mandatory inactive strategy. Every operating reward, project availability indicator, project cost, duration, generation law, verification law, and joint terminal belief-admission law factors through the current belief and the productive library state $K_L = (F_L, C_L)$. Raw insertion and compressed insertion obey

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (\text{S1.7})$$

We use the article's common discount factor $0 \leq \beta < 1$. The finite-horizon proof needs no contraction assumption; the stationary result uses the recorded contraction clause, which the finite Bellman specification below satisfies.

Calendar horizon and Bellman specification.

There are h operating reward dates remaining, $\bar{V}_0 = 0$, and no terminal salvage. A project may start only if $d_q \leq h$; there is no unfinished terminal project. Write $a_q \in \{0, 1\}$ for operation while active and $\kappa_q(b, K)$ for initiation cost. For $h \geq 1$, Continue has value

$$Q_h^0(b, K) = F(b) + \beta \sum_{b'} P_B(b, b') \bar{V}_{h-1}(b', K).$$

An available project that fits the horizon has value

$$Q_h^q(b, K) = -\kappa_q(b, K) + \sum_{\mathbf{b}, o} \Lambda_q(\mathbf{b}, o \mid b, K) \left[a_q \sum_{t=0}^{d_q-1} \beta^t F(b_t) + \beta^{d_q} \bar{V}_{h-d_q}(b_{d_q}, \text{addK}(K, o)) \right]. \quad (\text{S1.8})$$

The value $\bar{V}_h$ is the maximum over Continue and those projects. The raw recursion replaces K by L and addK by $L \oplus o$. The stationary operators use the same expressions with an unindexed continuation function and all available projects.

Theorem S1.1 (Raw-to-compressed controlled semi-Markov projection). *Under the preceding assumptions, (b, K_L) is a controlled semi-Markov state at embedded decision epochs for the raw library process. Raw and compressed finite-horizon values agree at*

every raw state, and the complete sets of maximizing actions agree under the natural action identification. Under the stationary discount clause, the raw and compressed Bellman operators have unique fixed points with the same values and maximizing stationary-action correspondence.

Proof Fix a raw library L , belief b , and feasible action. For Continue, the operating reward and belief transition depend on L only through F_L , hence through K_L . For a project q , the factorization assumption makes feasibility, cost, duration, the incumbent operating-reward block, and the joint completion law functions of (b, K_L, q) . Equations (S1.1)–(S1.4) show that this law is normalized without imposing independence between the belief path and admission. Pushing it through the deterministic raw update and using (S1.7) gives exactly the compressed transition law (S1.6). Its support consists of realizable compressed states. Thus two raw libraries in the same K -fiber have the same feasible actions, current reward blocks, durations, and conditional laws of the next compressed decision state.

For a finite calendar horizon h , prove equality of every corresponding action value by strong induction on h . The claim is immediate at horizon zero. Continue invokes the induction hypothesis at $h - 1$; a feasible project invokes it after its positive duration d_q , at $h - d_q < h$. Current rewards and transition laws already agree by the preceding paragraph. Each corresponding action value therefore agrees, as do the finite maxima and the complete maximizing-action sets. Finiteness makes every maximum attained.

For the stationary clause, cast the exact rational tables into the reals and use the sup norm on the finite state carrier. Conditional expectation is nonexpansive. Continue discounts continuation by β , and a project discounts it by $\beta^{d_q} \leq \beta$. Maximization over the common finite action set preserves this modulus, so both Bellman operators are β -contractions. Bellman intertwining, established by the action-value identity above, maps the compressed fixed point to a raw fixed point. Banach uniqueness gives value equality; the same pointwise action-value identity gives the complete maximizing stationary-action correspondence. At $\beta = 0$ both operators have modulus zero and the same argument applies. No closure detectability assumption enters this forward projection. For an arbitrary joint path–outcome law, a state observed during an unfinished project may need its initiation information or observed path history. This theorem makes no sufficiency claim for current belief, project identity, and remaining duration alone between decision epochs. $\square$

S2 Productive Preservation and the Frontier–Closure Quotient

For a finite library A containing the inactive strategy, write

$$F_A(b) = \max_{s \in A} j_s(b), \quad C_A = \text{cl} \left(\bigcup_{s \in A} \text{mods}(s) \right), \quad K_A = (F_A, C_A).$$

The forward preservation result uses only the process factorization and local update assumptions stated in Section S1. A converse needs a distinct detectability assumption: whenever two libraries have equal frontiers but unequal closures, at least one availability-tagged project cost, duration, incumbent-reward block, or joint terminal observation differs after the declared generation, verification, admission, and projection steps.

Theorem S2.1 (Frontier-closure characterization). *If $F_A = F_L$ and $C_A = C_L$, then A and L induce identical productive observations and identical finite-horizon values and maximizing actions. The stationary conclusions of Section S1 also agree under its discount clause. If closure is detectable in the preceding sense, equality of all productive observations together with frontier equality implies $C_A = C_L$.*

Proof Frontier-closure equality is $K_A = K_L$. The factorization assumption then identifies project availability, cost, duration, incumbent operating rewards, generation and verification laws, and the conditional joint completion law. The normalized admission map and deterministic pushforward from Section S1 therefore identify the complete productive transition observations. The finite-horizon and stationary value and action claims follow from Theorem S1.1. Conversely, suppose the frontiers and all named productive observations agree. If the closures differed, detectability would produce a named observation that differs, a contradiction. Hence the closures agree. Detectability is used only for this reverse implication. $\square$

Corollary S2.2 (Source-relative safe compression). *For $s_0 \in A \subseteq L$, exact equality $F_A = F_L$ and $C_A = C_L$ is sufficient for productive preservation. With exact positive active weights and $w_{s_0} = 0$, the finite safe-feasible family contains L and therefore has at least one minimum-burden member.*

Proof The productive claim is the forward direction of the theorem. The source is safe relative to itself. A finite source has finitely many sublibraries, so the nonempty feasible set has an attained minimum under exact weights. $\square$

The converse characterization is deliberately restricted. It is not a claim that K is a generic coarsest bisimulation or a full-abstraction theorem. It states that the declared finite process factors through K , and that the additional detectability clause is sufficient to recover closure equality from the named productive observations.

The operating observation is the maximum modeled payoff of one catalog entry. An entry may already specify a portfolio, but the result does not preserve an unmodeled allocation across several entries. Such an application requires a preservation target containing the relevant joint payoff information and feasible allocations before the factorization argument can be applied.

S2.1 Operational factorization audit

The assumption can be checked against a proposed application by asking whether, conditional on belief and active-project state, every operating reward, generation law, verification outcome, admission probability, duration law, completion observation, and next-library update depends on the raw library only through (F_A, C_A) . Provenance, cardinality, identity-specific switching cost, or a requirement for a named parent violates the assumption unless it is added to the compressed state. A finite module-gated construction satisfies the assumption when all projects are indexed only by requirements contained in C_A , incumbent rewards use only F_A , and all subsequent kernels use only belief, active-project variables, and those two objects.

The following exact countermodel shows why the checklist is substantive. Let s_0 be inactive and let active strategies a and b have identical zero profiles and the same single module. Then libraries $\{s_0, a\}$ and $\{s_0, b\}$ have identical frontier and closure. If a project is available only when the raw identity a is present, their project menus differ. The model therefore does not factor through (F, C) , and the projection theorem makes no claim about it. The countermodel changes no theorem: it falsifies an omitted assumption, not the stated conditional result.

S3 Tagged Cover, Complexity, and Boundary Counterexamples

This section gives the complete proofs supporting the article's identity-closure covering representation and complexity result. The semantic safe-feasibility condition is defined for arbitrary declared closure; only the tagged row reduction and the algorithms built on it assume identity closure.

S3.1 An exact tagged covering representation under identity closure

Fix a finite belief set B , a finite module carrier M , a finite strategy catalog S , and a source library $L \subseteq S$. Every admissible library contains the distinguished inactive strategy s_0 . Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$, $j_{s_0}(b) = 0$ for every b , and $\text{mods}(s_0) = \emptyset$. Each $\text{mods}(s)$ is a subset of M . For an admissible library A , let

$$F_A(b) := \max_{s \in A} j_s(b), \quad U_A^{\text{raw}} := \bigcup_{s \in A} \text{mods}(s), \quad C_A := \text{cl}(U_A^{\text{raw}}).$$

The maximum is attained because A is finite and nonempty. The inactive strategy also implies $F_A(b) \geq 0$.

In this subsection only, closure is the identity on 2^M . Thus $C_A = U_A^{\text{raw}}$ for every A . Define the source-relative tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}. \quad (\text{S3.1})$$

The constructors are disjoint: a frontier tag and a module tag are different even if their underlying identifiers have the same printed label. For each source strategy $s \in L$, define

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (\text{S3.2})$$

Equality in the first set is exact. Consequently every tied source maximizer covers the frontier row; no tie-breaking rule or distinguished maximizer enters the construction.

Theorem S3.1 (Tagged-cover equivalence under identity closure). *Under the assumptions above, let L' be an admissible source sublibrary: $s_0 \in L' \subseteq L$. Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (\text{S3.3})$$

All belief rows are retained in (S3.3). If $F_L(b) = 0$, the mandatory inactive strategy attains that frontier and automatically covers $\text{frontier}(b)$.

Proof Suppose first that $F_{L'} = F_L$ and $C_{L'} = C_L$. Fix $b \in B$. The finite maximum defining $F_{L'}(b)$ has an attainer $s \in L'$. Frontier equality gives $j_s(b) = F_{L'}(b) = F_L(b)$, so $\text{frontier}(b) \in \mathcal{R}_s$. This argument is existential and therefore permits any number of tied attainers. Next fix $m \in C_L$. Identity closure and closure equality give

$$m \in C_{L'} = U_{L'}^{\text{raw}} = \bigcup_{s \in L'} \text{mods}(s).$$

Hence some $s \in L'$ carries m , so $\text{module}(m) \in \mathcal{R}_s$. Every tagged requirement is covered.

Conversely, suppose every tagged requirement is covered. Because $L' \subseteq L$, maximization over the smaller set gives $F_{L'}(b) \leq F_L(b)$ for every b . Coverage of $\text{frontier}(b)$ supplies some $s \in L'$ with $j_s(b) = F_L(b)$. Therefore $F_L(b) = j_s(b) \leq F_{L'}(b)$, proving frontier equality pointwise. This also covers the zero-frontier case: $s_0 \in L'$ is itself a valid attainer.

For modules, source inclusion implies $U_{L'}^{\text{raw}} \subseteq U_L^{\text{raw}}$. If $m \in U_L^{\text{raw}} = C_L$, coverage of $\text{module}(m)$ supplies a retained carrier, so $m \in U_{L'}^{\text{raw}}$. Thus the two raw unions are equal. Identity closure converts this equality into $C_{L'} = C_L$. A module may have one or many source carriers; the row constraint requires at least one and does not select a canonical carrier. $\square$

Weighted binary formulation.

Let the exact rational retention weights satisfy $w_{s_0} = 0$ and $w_s > 0$ for $s \neq s_0$. Introduce one variable $x_s \in \{0, 1\}$ for each $s \in L$, where $x_s = 1$ means that s is retained. The theorem gives the exact formulation

$$\min_x \sum_{s \in L} w_s x_s, \quad (\text{S3.4})$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b) = F_L(b)} x_s \geq 1, \quad b \in B, \quad (\text{S3.5})$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (\text{S3.6})$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (\text{S3.7})$$

The source-sublibrary restriction is built into the column set L . The selected library is $L_x = \{s \in L : x_s = 1\}$, its burden is exactly the objective in (S3.4), and Theorem S3.1 identifies its feasible vectors with the exact innovation-safe sublibraries. Zero-frontier rows are redundant after $x_{s_0} = 1$, but retaining them makes the representation a literal tagged copy of every declared belief and avoids a hidden positive-frontier convention.

Why arbitrary closure is different.

For a nonidentity closure operator, equality of closed unions need not be equivalent to retaining a raw carrier for each member of the source closure. Capabilities may be generated jointly. A correct general-closure model must enforce the set-valued equality

$$\text{cl}\left(\bigcup_{s: x_s=1} \text{mods}(s)\right) = C_L$$

or use a separately proved valid extended formulation. The independent module rows in (S3.6) have no automatic validity in that setting.

Exact boundary examples.

The following finite examples use integer profiles and exact finite sets.

1. *Frontier rows alone are insufficient.* Take one belief, one module m , identity closure, and $L = \{s_0, s_m\}$, with both profiles zero, $\text{mods}(s_0) = \emptyset$, and $\text{mods}(s_m) = \{m\}$. The sublibrary $\{s_0\}$ covers the only frontier row and preserves the zero frontier, but its closure is empty rather than $\{m\}$.
2. *Module rows alone are insufficient.* Take one belief, identity closure, and $L = \{s_0, s_1\}$, with $j_{s_0} = 0$, $j_{s_1} = 1$, and both module sets empty. The source module row family is empty, so $\{s_0\}$ satisfies it vacuously, but its frontier is zero rather than one.
3. *Independent raw-module rows can fail under closure complementarity.* Let $M = \{a, b, c\}$ and

$$\text{cl}(A) = \begin{cases} A \cup \{c\}, & \{a, b\} \subseteq A, \\ A, & \text{otherwise.} \end{cases}$$

This map is extensive, monotone, and idempotent. Let the zero-profile source be $L = \{s_0, s_a, s_b\}$, where s_a carries only a and s_b carries only b . Then $C_L = \{a, b, c\}$, and the source is of course safe-feasible relative to itself. Nevertheless a raw-carrier row for $c \in C_L$ has no covering source strategy. The naive independent-row model therefore rejects even the source library.

The deterministic machine-readable realization is `journal/aor/fixtures/tagged_cover_equivalence_v1.json`. It records exact finite computation in Julia, including all sublibraries of its declared small-library grid; it is neither an exhaustive proof over arbitrary finite instances nor a solver optimality certificate. The universal core biconditional and exact burden identity are separately checked in the Lean source `TaggedSafeCompression.lean` under `formal/StrategyInnovation/Optimization/`. Those declarations formalize neither a complexity result nor an approximation guarantee.

S3.2 Computational complexity of exact compression

We use an explicit finite encoding. A source instance lists a finite strategy library $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set $\mathcal{B}$, exact binary-rational profiles $j_s(b)$, a finite module universe $\mathcal{M}$, and the incidence rows $\text{mods}(s) \subseteq \mathcal{M}$. Strategy s_0 is the

mandatory inactive strategy, has profile zero at every belief, carries no modules, and has zero burden. Every active strategy has a positive binary-integer weight w_i , and the threshold B is a nonnegative binary integer. Closure is identity closure,

$$C(A) = \bigcup_{s \in A} \text{mods}(s).$$

The source frontier is $F_L(b) = \max_{s \in L} j_s(b)$, and the burden of a sublibrary is $W(A) = \sum_{s_i \in A, i \geq 1} w_i$. The decision problem **IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID)** asks whether there is an $A \subseteq L$ such that $s_0 \in A$, $W(A) \leq B$, $F_A = F_L$ at every listed belief, and $C(A) = C(L)$.

The hardness source below is a deliberately normalized version of weighted set cover. In **FULL-UNION WEIGHTED SET COVER (FU-WSC)**, the input is a nonempty finite universe U , an indexed finite family $(A_i)_{i=1}^n$ whose union is U , positive binary-integer costs c_i , and a nonnegative binary-integer budget B . The question is whether some index set $I \subseteq \{1, \dots, n\}$ satisfies $\bigcup_{i \in I} A_i = U$ and $\sum_{i \in I} c_i \leq B$. Encodings that fail these stated syntactic restrictions are rejected, so FU-WSC is a language rather than an unverified promise problem.

Lemma S3.2 (Normalized weighted set cover). *FU-WSC is NP-complete.*

Proof Input validity is checkable in polynomial time. A selected-index bit vector is a polynomial certificate: unioning its listed incidence rows and adding its binary-integer costs verifies feasibility in polynomial time. Thus FU-WSC is in NP.

For hardness, reduce the standard **SET COVER** decision problem, one of Karp's NP-complete covering problems (Karp 1972). Write an input as $(U, (A_i)_{i=1}^n, k)$, with nonnegative cardinality bound k . If $U = \emptyset$, output the fixed yes-instance consisting of universe $\{u\}$, the one set $\{u\}$, cost one, and budget one. If $U \neq \emptyset$ but $\bigcup_i A_i \neq U$, output the fixed no-instance with universe $\{u\}$, the one set $\{u\}$, cost two, and budget one. Otherwise retain the universe and family, assign every set unit cost, and set the budget to k . Every output has a nonempty universe, a full-union family, positive integer costs, and a nonnegative integer budget. The first branch preserves the fact that the empty universe is coverable; the second preserves the fact that an uncovered element makes the source instance a no-instance; and in the third branch cost is exactly selected-set cardinality. The case split, union check, and copied incidence representation have polynomial size and construction time. Hence the mapping preserves yes- and no-instances and proves NP-hardness. $\square$

Theorem S3.3 (Identity-closure safe-compression complexity). *ISC-ID is NP-complete. More specifically, it remains NP-complete on the subclass with exactly one belief, zero profiles for every active strategy, an inactive strategy with zero profile, zero burden, and no modules, and a source library whose active module rows have full union. The result holds even when every active weight is one. Consequently, minimum-burden identity-closure safe compression is NP-hard. Any broader safe-compression model class containing this identity-closure subclass is NP-hard; no membership claim for an arbitrary nonidentity closure operator follows without a polynomial closure-equality verifier in its declared encoding.*

Proof Membership in NP. Encode a proposed retained library by one bit per source strategy. Inactive retention and source inclusion are direct bit checks. Binary-integer addition checks $W(A) \leq B$. For every explicitly listed belief, exact rational comparison finds the source and candidate maxima and checks their equality. Under identity closure, bitwise union of the listed module-incidence rows checks $C(A) = C(L)$. Integer and rational operations have bit complexity polynomial in the explicit input length. Thus ISC-ID is in NP.

Reduction algorithm. Given a valid FU-WSC instance $(U, (A_i)_{i=1}^n, (c_i)_{i=1}^n, B)$, create one module m_u for each $u \in U$, one active strategy s_i for each set A_i , and one mandatory inactive strategy s_0 . Use one belief b_0 . Set

$$j_{s_i}(b_0) = j_{s_0}(b_0) = 0, \quad \text{mods}(s_i) = \{m_u : u \in A_i\}, \quad \text{mods}(s_0) = \emptyset.$$

Give s_i weight c_i , give s_0 weight zero, take the complete constructed library as the source, use identity closure, and copy the budget without alteration.

Polynomial construction size. The construction adds one module per universe element, one active strategy per set, one inactive strategy, and one belief. Its module-incidence table is the set-cover incidence table; the remaining profiles are $n + 1$ zero entries, and costs and budget are copied. A dense encoding has size $O(n|U|)$ plus the copied identifier and integer bit lengths, while a sparse encoding is linear in the copied incidence list plus this overhead. Either encoding is polynomial in the corresponding explicit source encoding.

Forward feasibility. Suppose I is a set cover with $\sum_{i \in I} c_i \leq B$. Retain $A = \{s_0\} \cup \{s_i : i \in I\}$. Both source and retained frontiers equal zero at b_0 , with s_0 itself attaining that value. Identity closure gives

$$C(A) = \bigcup_{i \in I} \text{mods}(s_i) = \{m_u : u \in U\} = \bigcup_{i=1}^n \text{mods}(s_i) = C(L),$$

where the last equality uses the full-union condition. The inactive strategy is retained, and $W(A) = \sum_{i \in I} c_i \leq B$, so A is a yes-certificate for the constructed ISC-ID instance.

Reverse feasibility. Suppose A is a safe constructed sublibrary within threshold, and let $I = \{i : s_i \in A\}$. Exact closure preservation and the empty inactive module row imply

$$\bigcup_{i \in I} \{m_u : u \in A_i\} = C(A) = C(L) = \{m_u : u \in U\}.$$

The module labels are in one-to-one correspondence with U , so $\bigcup_{i \in I} A_i = U$. Thus I is a set cover. Frontier preservation adds no hidden selection requirement because the mandatory inactive strategy already attains the common zero frontier.

Threshold preservation. The inactive strategy has zero burden, active weights are copied exactly, and the threshold is copied exactly. Therefore $W(A) = \sum_{i \in I} c_i$ for every selected index set I , not merely for optimal selections. Hence $W(A) \leq B$ if and only if the corresponding set cover has cost at most B . The reduction is a polynomial many-one reduction. Together with membership in NP and Lemma S3.2, this proves NP-completeness of the stated restricted subclass and therefore of ISC-ID. The normalization can use the fixed no-instance with one singleton of cost one and budget zero in its uncovered-universe branch. It then uses unit costs in every branch, so the restricted subclass remains NP-complete with unit active weights. Solving the associated minimum-burden problem would decide whether its optimum is at most B , which proves NP-hardness of the optimization problem. $\square$

The encoding, normalization, and both directions above establish the human proof. Exact reduction fixtures are supplementary finite checks.

S4 Exact Preprocessing Propositions

Fix an identity-closure tagged covering instance after all declared mandatory columns have been selected. Let Q be the remaining requirement rows, A the available nonfixed columns, $R_s^Q = R_s \cap Q$, and let each optional weight be a strictly positive exact rational. There are no identity-sensitive side constraints, groups, exact-cardinality constraints, or mutual exclusions: this is the pure covering model. Every transformation records original identifiers, forced selections, fixed burden, equal-weight substitutions, and merged-row memberships.

Proposition S4.1 (Mandatory carrier and forced propagation). *If $u \in Q$ has exactly one carrier $s \in A$, every feasible completion selects s . Fixing s , adding w_s to the objective offset, deleting the rows it satisfies, and recomputing carriers preserves feasibility, the exact optimum value, every optimum identity, and reconstruction.*

Proof Any feasible completion must cover u , and s is its only available carrier, so s occurs in every completion. Removing already satisfied rows does not change the constraints on the remaining columns. Splitting every feasible objective into the fixed term w_s and its residual term is a bijection between feasible residual selections and feasible original selections containing the already forced columns. Repeating the argument after every selection proves propagation. $\square$

Proposition S4.2 (Coverage dominance). *Suppose $s, t \in A$, $s \neq t$, $R_s^Q \subseteq R_t^Q$, and $w_t \leq w_s$. If s is not mandatory, deleting s preserves feasibility, the optimum value, and at least one optimum. If $w_t < w_s$, no optimum contains s , so all optimizer identities are preserved. If $w_t = w_s$, all optimizer identities need not be preserved; a reconstruction record must retain the admissible substitution $s \leftrightarrow t$ where its surrounding selection remains feasible.*

Proof Take a feasible selection containing s . If it also contains t , then deleting s preserves coverage and strictly lowers burden because $w_s > 0$. If it does not contain t , replace s by t . Superset coverage preserves feasibility and the burden weakly falls. Thus every feasible solution has a no-worse representative without s , which proves feasibility and optimum-value preservation and supplies at least one reduced optimum. Under strict inequality, either case strictly improves any selection containing s , so no original optimum uses it. Under equality, replacement can exchange two distinct optimizer identities without changing burden. $\square$

Proposition S4.3 (Duplicate columns and empty contribution). *For columns with identical residual coverage, retaining a stable minimum-weight representative and deleting the others preserves feasibility, the exact optimum value, and at least one optimum. Strictly heavier duplicates cannot occur in an optimum. Equal-weight duplicates require an explicit substitution class to reconstruct all optimizer identities. A nonmandatory positive-weight column with empty residual coverage may be deleted; it occurs in no optimum.*

Proof Identical-coverage replacement leaves every row status unchanged and weakly reduces burden, with strict reduction for a heavier duplicate. Stable choice only fixes which representative remains in the reduced model. Recording the equal-weight class recovers the alternative identities consistent with the rest of a lifted selection. An empty-contribution column changes no residual constraint, and deleting it strictly lowers any selection that contains it. $\square$

Proposition S4.4 (Duplicate requirement rows). *If two residual requirement rows have identical carrier sets, replacing them by one representative preserves feasibility, optimum value, at least one optimum, and all optimizer identities. Recording both original row identifiers preserves the feasibility certificate.*

Proof A selection intersects one carrier set if and only if it intersects the identical other carrier set. The two inequalities therefore define the same feasible selections and do not enter the objective. Merging them changes no optimizer; the row-membership record merely expands the final certificate. $\square$

The implementation applies the rules to a stable fixed point and restarts its ordered scan after every transformation. Each nontrivial step removes a row or column or fixes a previously available column, so finiteness proves termination. The complete machine audit distinguishes forced columns, dominated columns, duplicate columns, empty columns, satisfied rows, and duplicate rows. These are exact combinatorial propositions. Julia randomized exhaustive tests and Lean finite-mask identities are separate evidence layers; neither turns a downstream solver status into a formal optimality proof.

S5 Algorithm Proofs, Certificates, and Pseudocode

The following proofs concern validated tagged identity-closure instances. Every returned library is reconstructed to original identifiers and then checked against the original source frontier, source closure, mandatory set, and exact burden. Approximation statements do not extend to arbitrary closure.

S5.1 Independent exact algorithms

The tagged representation gives two independent exact finite methods. They share only validated input, mandatory-selection preprocessing, and the final source-library certificate; their search procedures are different. Complete enumeration is the small-library oracle. Requirement-mask dynamic programming is parameterized by the number of residual tagged requirements.

Fix every mandatory strategy first. Let W_0 be their exact burden, delete every requirement they satisfy, and let the residual instance have strategies $1, \dots, n$ and requirements $1, \dots, r$. Write $R_i \subseteq [r]$ for the residual tagged coverage of strategy i and $w_i > 0$ for its exact weight. Mandatory preprocessing alone removes no optional strategy, so it preserves all optimizer identities. If a serialized instance already contains a valid stronger preprocessing map, the algorithms use its residual model, exact offset, and declared reconstruction map; complete-tie output is rejected when that map says that some equal-cost identities are unreconstructable.

The preprocessing assumptions and reconstruction qualifications are proved in Section S4.

S5.1.1 Complete enumeration

The enumeration method visits every one of the 2^n residual binary selections. It tests tagged coverage directly, sums burden in exact rational arithmetic, and retains either every minimum or the declared stable representative. It then lifts forced selections and rechecks mandatory retention, the original operating frontier, the original identity closure, and the exact burden independently of the search test.

Proposition S5.1 (Correctness of complete enumeration). *On a valid finite tagged identity-closure instance, complete enumeration returns a minimum-burden safe source sublibrary. When complete-tie retention is requested and the preprocessing map preserves all optimizer identities, it returns every minimum-burden source selection.*

Proof Under mandatory-only preprocessing, every admissible source sublibrary is the forced set together with exactly one of the 2^n residual subsets. Enumeration evaluates all of them. Its feasibility test is therefore true for precisely the feasible residual subsets, and its exact minimum is the residual optimum. Adding the selection-independent burden W_0 preserves objective order, and lifting is a feasibility- and burden-preserving bijection.

If the input already carries stronger preprocessing, the preprocessing propositions establish optimum-value preservation and a valid canonical lift; the solver does not reassert a bijection for eliminated columns. When the map also declares all optimizer identities reconstructable, its equal-coverage classes enumerate the additional optimal identities. Finally, the independent original-library checks exclude a corrupted lift or incidence-only certificate. The tagged-cover equivalence theorem identifies every checked lift with a safe source sublibrary. $\square$

S5.1.2 Requirement-mask dynamic program

Encode a requirement subset by an r -bit mask; union is bitwise OR. The implemented forward recurrence is the following.

1. Set $D_0[0] = 0$ and leave every other mask unreachable.
2. For $i = 1, \dots, n$ and every mask q , carry the skip transition $(q, D_{i-1}[q])$ and the take transition $(q \text{ OR } R_i, D_{i-1}[q] + w_i)$ into layer i , keeping the lower exact burden.
3. At equal burden, keep the first predecessor under the declared stable traversal, or retain every tied predecessor when complete-tie mode is requested.
4. Return the entry for the all-ones mask, add W_0 , reconstruct the original selection, and run the independent source-library certificate.

The implementation scans masks in increasing integer order and strategies in canonical identifier order, emitting the skip transition before the take transition. In representative mode it keeps the first exact predecessor; in complete-tie mode it stores the complete optimal predecessor DAG and expands it only during reconstruction. Thus iteration and representative tie selection are deterministic; hash-table iteration order does not determine output.

Theorem S5.2 (Exact requirement-mask dynamic programming). *For a valid finite tagged identity-closure instance, the recurrence above returns a minimum-burden safe source sublibrary and reconstructs at least one optimizer. In representative mode it uses $O(n2^r)$ exact-arithmetic and mask operations and $O(2^r)$ cost states plus $O(n2^r)$ reconstruction-aware predecessor records. If only optimum value were required, the two cost layers would use $O(2^r)$ memory. Complete-tie retention is output-sensitive: materializing T tied reconstruction vectors adds $O(Tn)$ bits and can be exponential in n .*

Consequently, the representative algorithm is fixed-parameter tractable in r : its bit-complexity is 2^r times a polynomial in the explicit input length. It is not polynomial in the full input size when r is unrestricted.

Proof For $i \in \{0, \dots, n\}$ and mask q , let $D_i[q]$ be the minimum exact burden of a subset of the first i residual strategies whose coverage union is exactly q , with $D_i[q] = +\infty$ if no such subset exists. We prove the stored invariant by induction on i . At $i = 0$, only the empty subset exists, so $D_0[0] = 0$ and all other states are unreachable.

Assume the invariant at $i - 1$. Every subset of the first i strategies either omits strategy i , in which case it supplies the skip candidate already characterized by D_{i-1} , or contains strategy i . Removing i from the latter subset leaves some predecessor mask q and cost at least $D_{i-1}[q]$. Replacing that predecessor by a minimum-cost one and reinserting i produces the same mask q OR R_i at cost $D_{i-1}[q] + w_i$. Conversely, every skip or take transition constructs a subset of the first i strategies with the stated mask and burden. Taking the minimum over these exhaustive and disjoint cases proves the invariant for layer i . Stored predecessor records follow the same transitions, so every finite state has a valid path to the base layer. Equal-cost merging preserves the first predecessor under the declared deterministic traversal; retaining every equal-cost predecessor preserves the complete optimal predecessor DAG. Expanding that DAG preserves every tied residual identity, subject to the explicitly enforced output cap.

At $i = n$, a residual selection is feasible exactly when its union is the all-ones mask. Hence $D_n[2^r - 1]$ is the residual optimum. Adding the fixed mandatory offset and lifting preserve objective value directly under mandatory-only preprocessing; for a supplied stronger map, those properties are the content of the preprocessing propositions. Applying the tagged-cover equivalence proves safe-compression optimality. The independent exact certificate rechecks the source frontier, source closure, mandatory strategies, and burden rather than using the DP mask as its sole evidence.

There are 2^r masks and two relaxations per strategy-mask pair, giving $O(n2^r)$ recurrence operations in the unit-cost mask/arithmetic model. Two layers contain at most 2^r exact costs. The implementation additionally stores optimal predecessor records across the n layers, giving the stated reconstruction-aware bound. Exact rational sums and comparisons have bit cost polynomial in the explicit input bit length: sums of at most n encoded weights have polynomially bounded numerator and denominator length. Thus the bit-time bound is $2^r \text{ poly}(|I|)$, which is fixed-parameter tractable in r and remains exponential when r is part of unrestricted input. $\square$

S5.1.3 Evidence boundary and counters

Both methods are exact finite computations, not Lean kernel verification and not mixed-integer solver evidence. The supporting mask-union declaration does not change that status. The standard result object records the exact objective and checked selections separately from wall-clock measurements. For enumeration, candidate count

is the number of binary subsets tested. For the DP, a visited state is a reachable mask at one strategy layer and a transition is one evaluated skip or take relaxation; the final reachable-state count is reported separately. Runtime fields are single-run measurements split into preprocessing, search, and reconstruction/certification scopes. They are instrumentation, not an algorithmic scaling finding; scaling evidence requires the separately locked benchmark design.

S5.2 Approximation-guaranteed greedy construction

The identity-closure reduction also permits a fast constructive method with a classical performance guarantee. Fix all mandatory strategies, including the inactive strategy, and remove the tagged requirements they cover. Let U^0 be the residual requirement universe, let $W_0 \geq 0$ be the fixed mandatory burden, and let each residual strategy s have positive exact weight w_s and residual coverage $R_s \subseteq U^0$. For nonempty U^0 , write

$$d = \max_s |R_s|, \quad H(d) = \sum_{i=1}^d \frac{1}{i}.$$

The weighted rule is the standard set-cover greedy algorithm of [Chvátal \(1979\)](#), applied only after the semantic reduction:

1. initialize the selected set with every mandatory strategy and set $U \leftarrow U^0$;
2. while $U \neq \emptyset$, choose a residual strategy minimizing

$$\frac{w_s}{|R_s \cap U|}$$

among strategies with nonzero new coverage, breaking an exact tie by the lowest original strategy index in canonical identifier order;

3. add the chosen strategy and replace $U \leftarrow U \setminus R_s$;
4. optionally examine the chosen nonmandatory strategies in reverse selection order and delete one exactly when complete tagged coverage remains.

All scores and comparisons in the implementation use `Rational{BigInt}`. A separate cardinality variant replaces w_s in the score by one, equivalently choosing the largest number of newly covered requirements. That variant has the burden guarantee below only when all residual optional weights are equal; with unequal weights, no approximation guarantee for the actual retention burden is asserted.

Theorem S5.3 (Transferred weighted-greedy guarantee). *Suppose the source library is finite, closure is identity closure, every residual tagged requirement has a carrier, mandatory weights are nonnegative, and every nonmandatory weight is positive. If $U^0 \neq \emptyset$, the weighted greedy construction returns an innovation-safe source sublibrary of burden W_G satisfying*

$$W_G \leq H(d) \text{ OPT},$$

where OPT is the minimum total burden of any safe source sublibrary and $d = \max_s |R_s|$. Deterministic reverse deletion returns burden $W_{GR} \leq W_G$ and hence

satisfies the same bound. If U^0 is empty, the mandatory library is optimal and the factor is one.

Proof First suppose U^0 is nonempty. The residual instance is an ordinary positive-cost weighted set-cover instance. Chvátal’s theorem applies to the rule minimizing cost per newly covered element and gives

$$G \leq H(d) \text{OPT}_R,$$

where G and OPT_R are, respectively, the greedy and optimal residual costs. The theorem permits any choice among exact minimum-ratio ties, so the canonical deterministic tie rule does not change the bound.

Every admissible source selection contains the mandatory strategies, and mandatory-only preprocessing removes no optional strategy. Thus $W_G = W_0 + G$ and $\text{OPT} = W_0 + \text{OPT}_R$. Since $W_0 \geq 0$ and $H(d) \geq 1$,

$$W_G = W_0 + G \leq W_0 + H(d) \text{OPT}_R \leq H(d)(W_0 + \text{OPT}_R) = H(d) \text{OPT}.$$

The tagged-cover equivalence then converts residual complete coverage plus mandatory retention into exact preservation of the source operating frontier and source identity closure.

Reverse deletion accepts a removal only after complete tagged coverage is rechecked. It never removes a mandatory strategy, and every removed optional strategy has positive weight. Therefore each accepted deletion strictly lowers burden, every rejected deletion leaves burden unchanged, and $W_{GR} \leq W_G$. Moreover, one reverse pass is enough for inclusion-wise irreducibility: if a strategy cannot be removed at its examination, later deletions cannot restore any missing coverage. The same $H(d)$ bound follows immediately. If U^0 is empty, mandatory retention is already feasible and every optional positive-weight strategy is unnecessary, so the mandatory library is the exact optimum. $\square$

A primary-source near-tight fixture.

Chvátal’s example has m requirements, singleton set $\{j\}$ of cost $1/j$ for each $j = 1, \dots, m$, and a set covering the entire universe with cost $1 + \epsilon$, where $\epsilon > 0$ and $1 + \epsilon < H(m)$. Greedy selects the singletons in order $m, m-1, \dots, 1$: when q requirements remain, singleton q has score $1/q$, whereas the universe set has score $(1 + \epsilon)/q > 1/q$. Its burden is $H(m)$, while the universe set alone is optimal with burden $1 + \epsilon$. The ratio $H(m)/(1 + \epsilon)$ tends to $H(m)$ as $\epsilon \downarrow 0$. The journal fixture realizes each requirement as an identity-closure module tag and lets the mandatory zero-weight inactive strategy cover the sole zero-frontier tag. Reverse deletion removes no singleton from the greedy endpoint.

Example S5.4 (Shared three-row greedy arithmetic). For $m = 3$ and $\epsilon = 1/12$, the singleton burden is $1 + 1/2 + 1/3 = 11/6$, the bundle burden is $1 + 1/12 = 13/12$, and the exact ratio is $22/13$. The Julia–Lean shared fixture checks only this arithmetic; it does not formalize Chvátal’s theorem or the transfer to safe compression.

Limitations and evidence boundary.

The guarantee is inherited set-cover theory; the contribution here is its proved transfer through the source-relative frontier-closure representation and exact postchecked implementation. The theorem does not apply to arbitrary nonidentity closure, because

independent module rows need not represent closure complementarities. It does not give the backward heaviest-safe-first deletion heuristic an approximation guarantee, and it does not make cardinality greedy a weighted-burden approximation when weights differ. The returned certificate is exact finite computation: it rechecks mandatory retention, tagged coverage, the original frontier, the original identity closure, and burden. It is neither exhaustive optimality evidence, mixed-integer solver evidence, nor Lean kernel verification.

S5.3 Certified rechecked-deletion portfolio

Forward greedy constructs a cover from the mandatory library. Rechecked deletion instead starts from the source and removes one entry at a time. The common implementation performs a complete fresh scan after every accepted deletion. For each retained nonmandatory strategy, it constructs the trial library and recomputes exact source-frontier equality and exact source-closure equality from the original profiles and module memberships. A deletion is accepted only after both equalities, mandatory retention, tagged coverage, and burden reconciliation pass. All other certificates are then discarded.

The deterministic portfolio contains heaviest-safe-first, lightest-safe-first, maximum immediate burden release, minimum remaining unique-carrier exposure, and declared source order. Under additive burden, the release from deleting one strategy is exactly its weight, so maximum immediate release and heaviest-safe-first are two names for the same ordering on this model; they are not treated as independent evidence.

For current retained set S , safe deletion candidate s , fixed tagged universe U_L , and tagged coverage R_t , define

$$E(s \mid S) = \left| \left\{ u \in U_L : \sum_{t \in S \setminus \{s\}} \mathbf{1}\{u \in R_t\} = 1 \right\} \right|.$$

The exposure rule minimizes $E(s \mid S)$, then maximizes exact immediate burden release, then uses the lowest canonical original strategy index. It uses only the source-defined incidence structure; it does not use dynamic value, held-out financial quality, realized return, or descendant outcomes. Heaviest and lightest rules break exact weight ties by lowest canonical original index. Declared-order deletion uses a complete caller-supplied permutation, or canonical original order by default.

The stochastic portfolio first draws a stable seeded permutation and then uses its earliest currently safe strategy at every fresh scan. Multi-start random deletion returns the lowest-burden certified endpoint across the declared starts; equal burdens are resolved by the lexicographically smallest selected-index tuple and then earliest start. The master seed, each derived seed, every random order, and every endpoint summary are retained.

Proposition S5.5 (Certified deletion endpoint). *Every algorithm in the rechecked-deletion portfolio returns a safe source sublibrary. Its reported endpoint is one-deletion irreducible after a complete exact final scan.*

Proof The source is safe relative to itself. At every accepted step, the trial library has been recomputed and certified to have the source frontier and source closure while retaining all mandatory entries. It is therefore safe, independently of the score used to choose it. Induction over the finite deletion trace gives safety of the endpoint. Every accepted deletion removes one retained nonmandatory strategy, so the trace terminates.

The final scan examines every retained nonmandatory strategy and records that its deletion fails exact source-relative feasibility. Hence no safe one-strategy deletion remains. Random ordering changes only the choice among already certified candidates and does not alter this argument. $\square$

The generic finite-trace safety implication in the first paragraph of the proof is Lean verified by `recheckedSafeDeletionEndpoint_feasible`. The declaration assumes a trace whose current-library redundancy certificates are already supplied. It does not verify the Julia scan, scoring, randomization, termination counters, or the reported final irreducibility scan; those clauses remain human proof plus exact Julia tests.

Safety and one-deletion irreducibility do not imply global resource efficiency. The singleton–bundle family in Theorem S6.1 remains a regression test: heaviest-safe-first and maximum immediate release return burden k while the global optimum has burden $1 + \epsilon$. Exact enumeration and mask dynamic programming, rather than any deletion status, supply small-instance optimum comparisons. Runtime counters are instrumentation for the separately registered computational study, not scaling evidence in this draft.

S6 Worst-Case Local–Global Distinction

This section proves the parameterized counterexample used to separate exact safety certification from optimization quality. It concerns the named heaviest-safe-first rule, not every deletion order.

S6.1 Worst-case gap for heaviest-safe-first deletion

Define HSF to recheck exact source-relative safety at the current library, delete a safely deletable active strategy of greatest weight, and repeat until none remains. Ties may be resolved by any fixed rule; the construction below has a unique greatest-weight choice at its decisive step. For a minimization instance with positive optimum, its output ratio is the retained burden divided by the global minimum safe burden.

Theorem S6.1 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Consider one belief, identity module closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy of zero weight, zero profile, and no modules. For each i , let s_i have zero profile, weight one, and module set $\{m_i\}$. Let the bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. At the full source library:*

1. *every active strategy is safely deletable, and s_B is the unique heaviest safely deletable strategy;*
2. *HSF deletes s_B and stops at the library of all k singletons, which is inclusion-wise irreducible and has burden k ;*

3. the bundle-only active library is the unique global minimum safe representation and has burden $1 + \epsilon$.

Consequently, the output ratio of HSF is

$$\frac{k}{1 + \epsilon}.$$

The rule HSF therefore has no constant approximation guarantee over finite identity-closure innovation-safe compression instances. Moreover, one-deletion irreducibility alone has no constant worst-case burden guarantee relative to the global safe optimum.

Proof All active profiles and the mandatory inactive profile are zero. The source frontier is therefore zero and remains zero in every admissible sublibrary. Under identity closure, exact safety in this family reduces to covering all of $m_1, \dots, m_k$. A retained active set covers those modules if and only if it contains s_B , or it contains every singleton $s_1, \dots, s_k$.

At the full source, removing any singleton leaves s_B , so every singleton is safely deletable. Removing s_B leaves all singletons, so the bundle is safely deletable as well. Because $\epsilon > 0$, the bundle weight $1 + \epsilon$ is strictly greater than each singleton weight one. Thus it is the unique heaviest currently safe strategy, and HSF deletes it first.

After that deletion, singleton s_i is the only retained carrier of m_i . Deleting it would change the closure, so no further deletion is safe. More strongly, every proper sublibrary of the singleton endpoint omits some s_i and hence loses m_i ; the endpoint is inclusion-wise irreducible. Additivity of burden gives endpoint burden k .

The bundle-only active library is safe and has burden $1 + \epsilon$. Any safe library without the bundle must retain all k singletons and has burden $k > 1 + \epsilon$. Any safe library with the bundle and at least one singleton has burden strictly above $1 + \epsilon$, since active weights are positive. The bundle-only library is therefore the unique global optimum, and division by its positive burden gives the stated ratio.

Finally, fix the rational value $\epsilon = 1/2$. It satisfies the family condition for every $k \geq 2$, while the ratio is $2k/3$, which exceeds any proposed finite constant for a sufficiently large integer k . This proves the two negative guarantees: the named rule reaches these endpoints, and the endpoints themselves are one-deletion irreducible. It does not say that every deletion rule reaches them. $\square$

Example S6.2 (Shared finite gap arithmetic). At $k = 4$ and $\epsilon = 1/2$, the heaviest-safe-first endpoint has burden 4, the bundle optimum has burden $3/2$, and their exact ratio is $8/3$. The common Julia–Lean fixture checks this displayed arithmetic and the Julia generator separately checks the corresponding finite safety trace. The Lean declaration does not formalize the universal safety characterization, the named algorithm, or the unboundedness argument.

The deletion-order boundary is exact. If a rechecked maximal deletion sequence removes the bundle first, each remaining singleton is essential and the sequence stops at burden k . If it removes any singleton first, the bundle becomes essential for that singleton’s module; every remaining singleton stays redundant while the bundle is retained, so maximal deletion ends at the bundle-only optimum. Thus the theorem applies to the specified heaviest-safe-first rule and supplies bad one-deletion-irreducible

endpoints, but it does not apply to every deletion order. In particular, it is not a lower bound for forward weighted-cover construction or for all certified deletion heuristics.

S7 Benchmark design and complete outcome accounting

The registered identity-closure benchmark contains 173 instances: 24 small exact, 128 structured, and 21 adversarial. The small family supports independent exact comparisons. The structured family varies source size, tagged requirements, incidence density, module overlap, bundles, unique carriers, weight dispersion, and frontier-module association. The adversarial family exercises the proved deletion gap, ties, duplicates, dominance, rare carriers, symmetry, and bundles. The instance registry fixes the realized comparison set; a requested but infeasible generator cell remains an unavailable instance.

S7.1 Methods and evidence

The method grid includes enumeration, mask dynamic programming, two registered HiGHS formulations, weighted and cardinality greedy, greedy followed by reverse deletion, deterministic deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate release and heaviest-safe-first are identical under additive weights and constitute one scientific method. The record keeps the two registered labels so that run-unit counts remain reproducible.

The independent audit checks original-coordinate mandatory retention, tagged coverage, complete frontier and closure equality, and exact burden. It accepts 3526 libraries among 3907 terminal run units. All 381 unsuccessful units remain in the denominator. Of 38 exact-capable instances, 31 have all required exact-method burdens available and equal, and 7 have incomplete comparisons. Equal burdens with different optimizer identities occur in 9 comparisons and are consistent with ties. Medium structured references are exactly rechecked HiGHS incumbents; solver-reported global optimality is distinct from the exact feasibility audit.

The six implementation-error statuses arise when enumeration exceeds its registered applicability cap. They supply no enumeration result. Generation failures cover 16 source cells and remove complete requested structural combinations. No failed instance is replaced. No terminal run is a time-limit outcome; therefore no timeout-gap distribution is estimated.

S7.2 Finite deletion-gap check

The following registered fixtures compare the observed burden ratio with the proved singleton-bundle family. They check the finite implementation, while the theorem establishes the unbounded family result.

S7.3 Interpretation and execution conditions

Full preprocessing commonly resolves structured instances completely. This supports the importance of forced carriers in this design but leaves limited evidence about

Table S1 All registered terminal run units. Solved categories sum to 3,526; every other category remains in the denominator of 3,907.

Terminal status	Run units
GENERATION_FAILED	352
IMPLEMENTATION_ERROR	6
MEMORY_LIMIT	14
NO_PRIMAL_CANDIDATE	1
PROCESS_INTERRUPTED	7
REJECTED_NONBINARY_SOLVER_CANDIDATE	1
SOLVED	3420
SOLVED_BY_EXACT_PREPROCESSING	106
Total	3907

Scale	Parameter	Reference evidence	Observed ratio	Theoretical ratio
small	$k = 4, \epsilon = \frac{1}{2}$	exact finite	$\frac{8}{3}$	$\frac{8}{3}$
medium	$k = 16, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{32}{3}$	$\frac{32}{3}$
large	$k = 64, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{128}{3}$	$\frac{128}{3}$

Table S2 Registered adversarial instances for heaviest-safe-first. Equality between observed and theoretical ratios is a synthetic fixture check of the human-proved family, not an empirical approximation guarantee.

difficult residual search. Multi-start completes all starts on 143 instances and improves on its first start on 96; successful sensitivity runs do not replace failed primary runs.

Execution used Julia 1.12.6 and HiGHS v1.15.1 on an Apple M1 Pro, with 8 concurrent one-thread processes. Times include compilation and worker contention, and the execution record identifies the source worktree. They are not steady-state speed estimates. The registered performance profile uses unlike success targets across methods. The registered HC3 models also omit within-instance dependence and cannot recover missing generator cells. Neither output supports the article’s comparative or inferential claims. The published comparison instead uses exact agreement, certified burden, preprocessing, and the complete status census.

The data files named in the reproduction manifest supply the complete instance registry, exact-agreement records, method-level burden summaries, preprocessing and multi-start summaries, and terminal records. The PDF reports aggregates; no source row is removed from those current study records.

S8 Financial and mechanism methods

The point-in-time portfolio study is retrospective development evidence. Its historical setting was known during design. Its sequential freezes protect the stated within-study information boundary, without making the historical sample an independent confirmatory test.

S8.1 Financial universe, strategy grammar, and timing

For origin year y , years $y - 5$ through $y - 3$ form the source library; years $y - 2$ through y provide compression profiles; year $y + 1$ supplies proposal estimates and freezes policy choices; year $y + 2$ evaluates those choices. No evaluation value enters any choice. The primary universe contains the 100 most liquid eligible US common equities at the origin. The separate ETF replication selects up to 50 eligible plain ETFs and requires at least 20. One ETF origin fails its universe gate and remains unavailable. Eligibility uses predecision data and does not require subsequent survival.

The 96-specification grammar crosses three signals, two filters, two holding horizons, two allocation rules, two gross exposures, and two exit rules. Signals use 20- or 60-session momentum or five-session mean reversion; filters are unrestricted entry or a 100-session trend filter. Allocation is equal weight or inverse trailing volatility. Exposure is subject to the frozen security cap and a 10% annual portfolio-volatility ceiling; unused exposure remains cash. Intended weights formed at close t first earn the return indexed $t + 2$. Observed delisting returns are earned by the effective position and followed by cash. A required missing return makes the path unavailable; it is never imputed as zero. The current grammar and burden registries fix the remaining rule values and identifiers.

Formation and compression each use 12 profile rows: three years crossed with 5- and 30-basis-point one-way costs and risk aversion 1 and 3. Each floating profile is rounded to the nearest 10^{-12} and represented as an exact rational for comparison. Thus frontier equality is exact for the recorded table, without treating measured returns as exact economic quantities. The source contains all formation-frontier carriers plus the first three formation-ranked carriers of each required capability. It is frozen before compression profiles are computed. Each cell has 31 declared capabilities, including atomic rules, interfaces, relevant point-in-time data, corporate actions, aggregation, and cost/capacity support.

S8.2 Burden, compression, and proposal sets

A strategy's burden is the exact integer sum of its components in the frozen burden registry. It is a validation-computation index, with zero burden for mandatory cash. The safe arm preserves the source frontier and full capability union. The comparator first solves frontier-only retention and then adds whole source entries in descending mean compression score, breaking ties by identifier, whenever they fit within the safe arm's realized burden cap. Every endpoint is independently checked in the unprocessed coordinates. 3 safe cells are resolved by exact preprocessing; 34 receive HiGHS OPTIMAL status and an exact feasibility/burden postcheck.

Proposal actions comprise the grammar candidates excluded from the source docket and enabled by the retained requirements. The comparator action set is nested in the safe set before reading proposal scores. The complete candidate ledger retains source-docket exclusions, missing capabilities, rejected candidates, cash, and unavailable cells. The analysis unit is the origin; candidate specifications and daily returns are not independent replications. Adjacent origins share formation and compression histories and should not be treated as independent market replications either. Origin summaries are descriptive; no cross-origin significance claim is made.

S8.3 Certainty equivalent and the registered selection screen

For gross daily portfolio return r_t , one-way turnover q_t , and net return $n_t = r_t - 0.0005q_t$, the primary annual certainty equivalent is

$$\text{CE} = 252 \bar{r} - 252 \overline{0.0005q} - \frac{1}{2} 3 252 \widehat{\text{Var}}(n),$$

with the registered sample-variance convention. The mean, cost, and risk penalty are reported separately. The economic adoption hurdle is 0.0025 in annual CE units.

Proposal uncertainty uses 5,000 paired circular moving-block bootstrap draws with 20-session blocks. A draw applies the same resampled session indices to each candidate and baseline. If δ_i is an estimated annual CE difference and se_i is the standard deviation of its bootstrap differences, the reference maximum for each draw uses $(\delta_i - \delta_i^*)/\text{se}_i$, with zero used when the standard error is zero. The 90th percentile of that maximum is the one-sided critical value c , giving the screening score $\ell_i = \delta_i - c \text{se}_i$.

The comparator selects cash unless an eligible candidate's score over cash exceeds the hurdle. The safe policy defaults to that exact choice. Because the comparator is selected from proposal data, the safe reference family includes every complete safe-candidate by complete comparator-candidate contrast, including cash. Replacement requires the corresponding score to exceed the hurdle. Family shrinkage ranks only candidates that pass this gate: its center is the family mean, its latent variance is the nonnegative part of between-path sample variance minus median squared standard error, and its reliability weights shrink each candidate toward that center. Identifier order resolves remaining ties.

Coverage boundary of this screen.

The implemented reference uses $\delta_i - \delta_i^*$, the reflection of $\delta_i^* - \delta_i$, with standard errors fixed across resamples. A conventional lower confidence construction instead calibrates the upper sampling error $\delta_i - \theta_i$; bootstrap multiple-testing results require a valid approximation to the joint error law (Romano and Wolf 2005). Reflection does not preserve a general asymmetric error law. For example, with one contrast, unit standard error, and exact centered error $E = Z - 1$, $Z \sim \text{Exp}(1)$, the 90th percentile of $-E$ is $1 + \log(0.9)$. Using it in a lower bound gives coverage $1 - e^{-2}/0.9 \simeq 0.8496$, not 0.90. This is an analytic counterexample to an unconditional guarantee, not an estimate of coverage in our financial sample. No finite-sample coverage validation or applicable asymptotic argument is supplied for this particular screen. Its 90% level is therefore a tuning parameter. The recorded policy and all its outcomes are reported

as implemented; they do not establish familywise error control. The separate synthetic learner below uses a scalar null-calibrated threshold and does not validate this financial bootstrap rule.

S8.4 Origin-level outcomes and robustness

All 19 common-equity pairs choose the same action, so their held-out contrasts are exactly zero. One of the 18 complete ETF pairs replaces the comparator and realizes a negative contrast. The complete origin-level record follows.

Table S3: Origin-level point-in-time held-out contrasts. CE differences are innovation-safe minus frontier-only. A dash marks the prespecified failed ETF universe gate; it is retained, not imputed.

Origin	Equity CE diff.	Same choice	ETF CE diff.	Same choice
2005	0.0000	yes	–	–
2006	0.0000	yes	0.0000	yes
2007	0.0000	yes	0.0000	yes
2008	0.0000	yes	0.0000	yes
2009	0.0000	yes	0.0000	yes
2010	0.0000	yes	0.0000	yes
2011	0.0000	yes	0.0000	yes
2012	0.0000	yes	0.0000	yes
2013	0.0000	yes	0.0000	yes
2014	0.0000	yes	0.0000	yes
2015	0.0000	yes	0.0000	yes
2016	0.0000	yes	-0.0390	no
2017	0.0000	yes	0.0000	yes
2018	0.0000	yes	0.0000	yes
2019	0.0000	yes	0.0000	yes
2020	0.0000	yes	0.0000	yes
2021	0.0000	yes	0.0000	yes
2022	0.0000	yes	0.0000	yes
2023	0.0000	yes	0.0000	yes

The prespecified cost, risk-aversion, and hurdle grid preserves common-equity choice equality. ETF differences remain at the same origin and disappear at higher costs. Leave-one-origin-out summaries aggregate frozen choices without reselection. Unregistered larger-universe and closure-sham variants remain unavailable. Forced-maximum and full-family search summaries are diagnostics and never feed back into selection. Capacity calculations score frozen choices and retain missing inputs, incomplete paths, and participation-cap breaches.

S8.5 Calibrated synthetic bridge experiment

The two libraries have equal current frontier and burden; only the safe one can run the specified descendant project. The parameters are $\beta = 0.99$, $\rho = 0.98$, $\pi = 0.90$, and $d = 1$, so success probability is 0.882. The net bridge margin is $M = \beta^d \rho^d \pi G - \kappa$. An oracle researches precisely when $M > 0$; the learner researches when its noisy estimate exceeds an independently calibrated threshold.

Calibration uses 2479 complete financial proposal paths and 623647 residual observations, after removing cross-strategy date means within cells. It estimates scale, lag-one dependence, and tail thickness. Simulated proposal noise follows a stationary AR(1) with variance-normalized Student innovations, 100 burn-in sessions, 756 proposal sessions, and annualization 252. A separate 4,096-world null simulation fixes the threshold at the greater of the 0.0025 hurdle and its 95th percentile, giving 0.027008117603021155. Research cost and powered margin are each twice that threshold.

The four margins are negative one threshold, zero, 0.0025, and positive two thresholds. Each has 4,096 test worlds. Common random numbers pair regimes within a world; calibration and test seeds are disjoint. Mean intervals use mean plus or minus 1.96 standard errors; adoption uses Wilson intervals. All four regimes and all worlds remain in the ledger. The primary powered-oracle gate requires a positive lower endpoint and inclusion of the analytic margin. The powered learner must have a positive lower endpoint and at least 90% adoption. Null and adverse point adoption must not exceed 6.5% and 1%, respectively.

The article reports the full four-regime table. Its null point adoption is 5.76%, while its Wilson upper endpoint is 6.52%. The low positive dose does not yield a positive learned interval; the powered one does. The powered margin is a designed dose, and the experiment supports its declared synthetic mechanism rather than a market premium. Structural assertions check frontier, burden, and bridge availability in every world.

S8.6 Input access

Public financial outputs contain aggregate results, structural counts, and cryptographic seals. Raw CRSP/WRDS rows, security identifiers, return paths, and selected identities derived from licensed returns are excluded. Full source-data replication requires independent licensed access. Public aggregate tables and the synthetic ledger have separate reproducible checks identified in the reproduction guide.

S9 Theorem and Evidence Correspondence

The table records the proof and verification scope of the results used in this paper. A supporting finite fixture never upgrades the status of a universal result, and solver evidence establishes none of the mathematical claims in this table.

Table S4: Evidence status of the theory and algorithm results.

Result	Verification status	Supporting evidence and boundary
Raw-to-compressed projection	Lean verified for the encoded finite model; human proof for the displayed Bellman specification	Exact admission and pushforward, finite-horizon value equality, and stationary lifting under supplied contraction obligations; duration and operation are project-only in that encoding. The hidden-state filter and arbitrary within-project Markov sufficiency are not formalized.
Sharp normalized bridge loss	Lean verified for the parameterized canonical construction; human proof for the general optional-project bound	The canonical reward equals its cap. Below the cap the optional value is the positive part of net project value; equality with the cap bound is not asserted.
Operational–generative decomposition	Human algebraic identity	Substitution of the defined research-option premium; it is not a positivity or market-performance theorem.
Tagged-cover equivalence and weighted objective identity	Lean verified	Human proof; exact Julia boundary fixtures. The Lean result covers the encoded universal biconditional and burden identity, not complexity or approximation.
Identity-closure NP-completeness	Human proof only	Exact reduction fixtures and Julia tests; no Lean complexity framework.
Unbounded heaviest-safe-first gap	Human proof only	Exact rational family fixtures and Julia tests; Lean checks one finite arithmetic case only.
Preprocessing propositions	Human proof only	Lean verifies selected finite feasibility cores; objective preservation, reconstruction, propagation, and fixed-point claims are exact fixtures and Julia tested, not formally verified.
Enumeration and requirement-mask DP correctness, complexity, and FPT status	Human proof only	Exact cross-method fixtures and Julia tests; Lean mask identities do not formalize the recurrence invariant or complexity bound.
Weighted-greedy transfer and reverse deletion	Human proof only	The external Chvátal theorem is not re-formalized; finite ratios are exact fixtures and Julia tested.

Result	Verification status	Supporting evidence and boundary
Rechecked-deletion feasibility	Lean verified for an abstract supplied safe trace; human proof only for concrete-suite termination and final-scan irreducibility	Julia tests cover scoring, stale-certificate invalidation, reproducible randomness, counters, feasibility, and irreducibility. No approximation quality follows.

The Lean axiom audits recorded for the tagged-cover and preprocessing declarations print only `propext`, `Classical.choice`, and `Quot.sound`. The shared Julia–Lean fixture prevents manual transcription of selected finite rational values. The reproduction manifest identifies the required formal sources under `formal/StrategyInnovation/Optimization/`, `formal/StrategyInnovation/Fixtures/`, and `formal/StrategyInnovation/Audit/`.

S10 Reproducing the paper and its evidence

The accompanying editable source package builds the article, this resource, and their combined PDF. Run `python3 build.py -output ./release` after installing Python with `pypdf`, a LaTeX distribution with `latexmk`, and Poppler. The builder checks references, embedded fonts, searchable text, page counts, and internal destinations in the combined file. Its source manifest and SHA-256 checksums identify every distributed input. The three article figures have editable TikZ/PGFPlots sources; `build_figures.py` derives their numeric inputs from the public evidence and independently recomputes the synthetic intervals from all test worlds. `verify_review.py` checks those inputs and the small analytic review fixtures. These commands perform no new financial experiment. The current package is available from the repository’s [versioned preprint release directory](#).

S10.1 Public evidence and source binding

The `evidence/` directory contains the complete benchmark terminal CSV, instance and seed registries, exact-agreement and method summaries; the financial study’s public aggregate evaluation manifest and burden/capability registries; and the calibrated synthetic experiment’s configuration, calibration, regime and seed registries, summary, and complete world ledger. The machine-readable `REPRODUCTION_MANIFEST.json` records original repository paths and SHA-256 hashes for these files and the selected scientific sources. Internal study identifiers locate the exact implemented experiments.

The repository source commit is `bb58f33fd42f82f51b2866b44a666c3094f2932a`. The document source archive is self-contained for PDF rebuilding and public evidence inspection. Full computational replication additionally requires the repository’s pinned Julia and Lean environments and the scientific source files identified in the manifest. File hashes, rather than a moving branch name, bind the computational inputs.

S10.2 Mathematical and algorithmic verification

The main article and Sections S1–S6 contain the human arguments. Selected Lean modules cover the projection, tagged-cover result, finite preprocessing cores, and supplied safe-deletion traces. The correspondence in Section S9 states which complete results remain human proofs. The reproduction guide gives focused Julia fixture checks and Lean module commands; a successful finite fixture is not a universal proof.

S10.3 Numerical reproduction

The benchmark’s saved terminal records retain unsuccessful and unavailable runs. The exact audit reads those outcomes and rechecks returned libraries; regenerating a benchmark is a separate computation whose machine timing may change. The financial tables can be checked against distributed public aggregates, but portfolio paths and policy selection require independent licensed data access under the specified staging contracts. The synthetic experiment can be rerun from its public calibration and fixed configuration; its calibration and test worlds are distinct. The public calibration values suffice for this rerun; recalibrating those values from source market returns requires the licensed paths.

The accompanying `README.md` provides commands for each of these tasks and distinguishes rebuilding documents, checking public records, and recomputing scientific outputs. No raw or row-level licensed observation is included in the distribution.

Generative-AI disclosure

OpenAI ChatGPT and Codex assisted with LaTeX drafting and editing, software review, and repository-based analysis and reproducibility tasks. The author remains solely responsible for the resource. No AI system is an author and no generative-AI image is used.

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